Shoaib Mohammed

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SUMMARY

Software Engineer with 4+ years of experience designing and operating infrastructure for latency-sensitive systems and ML pipelines. Proficient in building REST APIs and React-based frontends using TypeScript, Node.js, Python and AWS. Experienced in production-grade deployment of gRPC services and protobuf-based schemas. Proven track record in building scalable CI/CD workflows, optimizing Kubernetes environments, and deploying observability stacks to support distributed microservices. Proficient in Terraform's, and cloud-native tooling (Kubernetes, AWS, Helm, Prometheus, OpenTelemetry). Passionate about enabling developer productivity through platform abstraction and automation. **Open to relocating anywhere in the USA.**

WORK EXPERIENCE

Client: Simtec - (Remote) June 2024 - Present

Role: Software Engineer

- Increased user engagement by 30% by developing scalable, high-availability web apps with React.js, Next.js, and Node.js.
- Improved API efficiency and cut response times by 30% using PostgreSQL optimization and Redis caching.
- Designed and led cloud infrastructure for ML training and deployment pipelines, integrating observability with Prometheus and Grafana to support continuous training and production monitoring.
- Designed and implemented distributed systems and data pipelines, leveraging Kafka, Redis Streams, and PostgreSQL, ensuring scalable and efficient data flow for real-time processing and inference.
- Developed and deployed gRPC-based microservices using Node.JS and Typescript, with protobuf schemas enabling structured inter-service communication.
- Managed multi-cloud Kubernetes infrastructure (EKS/GKE/AKS), implementing blue-green rollouts and auto-scaling strategies for zero-downtime deployments and improved system reliability.
- Collaborated with ML and platform teams to build self-service infrastructure tools, reducing platform support tickets by 40% and accelerating developer onboarding.
- Tuned Linux kernel parameters and implemented CPU pinning strategies to optimize Kubernetes workloads, achieving a 30% P99 latency reduction.
- Tuned network adapters, CNI plugins, and load-balancing configurations across hybrid clusters to reduce packet loss and latency in on-prem and cloud environments, improving service reliability under high load.
- Built scalable CI/CD pipelines and reusable Terraform + Helm modules to automate AWS and Kubernetes infrastructure provisioning, reducing deployment time by 70% and eliminating 90% of manual configuration errors.
- Automated cloud cost optimization with right-sizing policies and scheduled resource shutdowns, saving $40K annually in operational costs
- Implemented infrastructure security best practices, including IAM hardening, Vault/AWS Secrets Manager integration, and automated patching to meet compliance standards.

Illinois Institute of Technology *(Graduate Assistant)* -**Chicago, IL** May 2023 - May 2024

- Training and Development of University chatbot(Talon the Scarlet Hawk) powered by Gecko and integrated with ChatGPT & web scraping capabilities to provide real-time answers to student queries.
- Implemented Natural Language Processing (NLP) techniques to enable the chatbot to understand and process natural language input, resulting in more accurate and contextually relevant responses.
- Curated and collected relevant datasets to train the chatbot, ensuring a diverse range of topics and queries commonly encountered by university students. Conducted fine-tuning of Talon-Scarlet Hawk bot on the university-specific dataset, tailoring the model to better understand and respond to university-related queries.
- Integrated observability tools (OpenTelemetry, ELK stack) into platform stack to provide deep insights into system health and troubleshoot issues faster.

Client: Cognizant Mar 2020 - July 2022

Role: Software Engineer(Full Stack)

- Contributed to development of backend REST services in Javascript and Node.js; introduced gRPC support for selected service endpoints.
- Developed test automation pipelines using Selenium and CI systems like Jenkins, improving release cycle confidence.
- Applied TDD for backend services and built SPAs using React and Angular for internal dashboards and admin tools.
- Participated in backend refactor initiatives using Spring Boot and Kotlin to improve maintainability and performance.
- Ensuring cohesive and visually appealing user interface across various platforms and contributing to the overall user experience and the success of Client products.
- Applied TDD principles in Java microservices to increase code quality and reduce production incident rate by 30%; collaborated with QA and release teams to ensure regression coverage.
- Collaborated with design and QA teams to ensure consistent UX across platforms; contributed to frontend CI pipelines and unit test coverage using Jasmine/Karma.

PROJECTS & PUBLICATIONS

App Builder -No code Platform - *NodeJS|React| Web Container | OpenAI* **GitHub** Dec 2024

- Built an AI-powered website builder platform using Node.js and Web Containers, enabling instant website creation with real-time preview and GPT-based content generation for enhanced user experience.

Metaverse - **Virtual Office Space** - *WS |Express | Monorepo| ORM | TDD* **Github** Nov 2024

- Developed a scalable virtual environment using Node.js and WebSockets, adopting Test-Driven Development (TDD) and building a modular event-driven backend with real-time movement and user interactions.

Task Automation Platform - *Node Js|Kafka |Postgres|Next JS* **Github** July 2024

- Developed an automation platform enabling users to create workflows and trigger actions based on events, enabling event-driven workflows across services using Kafka as a queue and the Outbox Pattern for reliable message delivery and Consistency.

Brain Tumor Detection - *Transfer Learning|CNN|ResNet-50|Inception-V3|VGG-16| Colab* **IEEEXplore** Jan 2022

- This pattern recognition problem for automatic detection of brain-tumor from MRI-images greatly saves the precious time of the radiologist. The accuracy of the VGG network achieves more than 98.16% compared to ResNet-50 and Inception-v3.

CERTIFICATION

- Cisco® Certified CCNAv7: Introduction to Networks
- Cisco® Certified CCNAv7: Switching, Routing, and Wireless Essentials
- Cisco® Certified CCNAv7: Enterprise Networking,Security, and Automation
- Cisco® Certified DevNet Associate

CORE COMPETENCIES

- Languages & Scripting : Typescript,Python, Go, Java, Bash
- Web Frameworks : HTML/CSS, React, Nextjs, Tailwind CSS, Nodejs, Springboot, Express, gRPC,Proto Buffs, Websocket,WebRTC, Web
- Build & CI/CD : Github Actions, Workflows, GitLab CI, Jenkins, TeamCity (Kotlin DSL),Maven
- Databases & Storage : PostgreSQL,Sqlite,MySQL Prisma, Redis, MongoDB, Oracle
- DevOps & Infrastructure : Kubernetes, Helm, Docker, AWS (EC2,Lambda, S3, EKS,IAM), GCP, Terraform, Ansible
- Monitoring & Observability : Prometheus, Grafana, ELK Stack, Datadog, Splunk, Newrelic, OpenTelemetry.
- Job Orchestration & Messaging : Kafka, Airflow, Event-driven Automation
- Performance & Latency Testing : perf, tcpdump, Stress-ng, Celery

EDUCATION

Illinois Institute of Technology, Chicago

Masters in Computer Science

Jawaharlal Nehru Technological University, Hyderabad,India

Bachelor of Technology Computer Engineering